**STUDYNEST | CLASS 11 PHYSICS**

# QUESTION PAPER 1

## Chapter 10: Thermal Properties of Matter

**Maximum Marks: 40 | Suggested time: 60 minutes**

Instructions: Answer all questions. Show calculation steps and units. Original practice paper, not an official examination.

### Section A 1-mark questions - 8 x 1 = 8 marks

1. What is temperature's role at equilibrium? [1]
2. Convert 27C to kelvin. [1]
3. Specific heat capacity? [1]
4. Heat for temperature change without phase change? [1]
5. Latent heat relation? [1]
6. Heat transfer through vacuum? [1]
7. Ideal gas equation? [1]
8. Newton's law of cooling? [1]

### Section B 2-mark questions - 6 x 2 = 12 marks

9. 2 kg warmed 5 K, c=400: Q? [2]
10. =1.2 x 10^-5,L=2m,DeltaT=50K: extension? [2]
11. 100C in kelvin? [2]
12. Heat capacity of .5 kg with c=800? [2]
13. Heat to melt .2 kg ice, L=3.34 x 10^5? [2]
14. Why is temperature constant during boiling? [2]

### Section C 3-mark questions - 4 x 3 = 12 marks

15. .5 kg water c=4200 heated 10C: heat? [3]
16. Rod L=2m,=1.5 x 10^-5,DeltaT=40K: expansion? [3]
17. Mix .2 kg at 80C and .3 kg at 20C, equal c, no heat loss: final T? [3]
18. Gas at fixed volume warms 300K to 450K; initial p=100kPa. Final p? [3]

### Section D 4-mark questions - 2 x 4 = 8 marks

19. Describe three heat-transfer modes. [4]
20. Melt .2 kg ice at 0C then warm water by 20K; L=3.34 x 10^5,c=4200. Total Q? [4]

**End of question paper.**

# FORMULA AND EQUATION REFERENCE

### Chapter-specific equations and relations

- Temperature conversion: T(K)=theta(deg C)+273.15.
- Heat for temperature change: Q=mc Delta T; heat capacity C=mc.
- Latent heat during phase change: Q=mL.
- Linear expansion: Delta L=alpha L Delta T; area expansion approx 2alpha A Delta T; volume expansion approx 3alpha V Delta T for isotropic solids.
- Calorimetry without heat loss: heat lost = heat gained; for equal specific heat, T_f=(m_1T_1+m_2T_2)/(m_1+m_2).
- Ideal gas equation pV=nRT; at fixed volume p/T=constant.
- Newton cooling (small temperature difference): rate of heat loss proportional to temperature excess.
- Heat-transfer modes: conduction (microscopic interactions), convection (bulk fluid motion), radiation (electromagnetic waves).

Use equations only when their assumptions apply; keep units consistent.

# STEP-BY-STEP SOLUTIONS

Concept questions are answered directly; numerical problems show the governing equation and substitution.

### Question 1

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: No net heat transfer between systems in thermal equilibrium.
**Final answer: No net heat transfer between systems in thermal equilibrium.**

### Question 2

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: 300.15 K, approximately 300 K.
**Final answer: 300.15 K, approximately 300 K.**

### Question 3

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Heat needed to raise 1 kg by 1 K.
**Final answer: Heat needed to raise 1 kg by 1 K.**

### Question 4

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Q=mcDeltaT.
**Final answer: Q=mcDeltaT.**

### Question 5

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Q=mL.
**Final answer: Q=mL.**

### Question 6

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Radiation.
**Final answer: Radiation.**

### Question 7

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: pV=nRT.
**Final answer: pV=nRT.**

### Question 8

Principle / definition: Identify the physical quantity or law asked for, then apply the chapter definition.
Apply it here: Rate is approximately proportional to excess temperature for small differences.
**Final answer: Rate is approximately proportional to excess temperature for small differences.**

### Question 9

Working: Q=mc Delta T=2(400)(5)=4000 J.
**Final answer: 4000 J.**

### Question 10

Working: Delta L=alpha L Delta T=(1.2x10^-5)(2)(50)=1.2x10^-3 m.
**Final answer: 1.2 x 10^-3 m.**

### Question 11

Working: T=theta+273.15=100+273.15=373.15 K.
**Final answer: 373.15 K.**

### Question 12

Working: Heat capacity C=mc=(0.5 kg)(800 J kg^-1 K^-1)=400 J/K.
**Final answer: 400 J/K.**

### Question 13

Working: For melting at constant temperature, Q=mL=0.2(3.34x10^5)=66,800 J.
**Final answer: 66,800 J.**

### Question 14

Working: During boiling, supplied energy is used as latent heat to change liquid into vapour by overcoming intermolecular attractions. Temperature stays constant until the phase change is complete.
**Final answer: Energy supplies latent heat for phase change.**

### Question 15

Working: Q=mc Delta T=0.5(4200)(10)=21,000 J.

**Final answer: 21,000 J.**

### Question 16

Working: Delta L=alpha L Delta T=(1.5x10^-5)(2)(40)=1.2x10^-3 m.

**Final answer: 1.2 x 10^-3 m.**

### Question 17

Working: Heat lost=heat gained: 0.2c(80-T)=0.3c(T-20). Cancel c: 16-0.2T=0.3T-6; 22=0.5T; T=44 deg C.

**Final answer: 44C.**

### Question 18

Working: At fixed volume, p/T is constant. T_2/T_1=450/300=1.5, so p_2=100 kPa(1.5)=150 kPa.

**Final answer: 150 kPa.**

### Question 19

Working: Conduction transfers energy through microscopic interactions without bulk movement; convection transfers energy by bulk flow in fluids; radiation transfers energy by electromagnetic waves and can occur through vacuum.

**Final answer: Conduction via microscopic interactions; convection by bulk flow; radiation via electromagnetic waves.**

### Question 20

Working: First melt the ice: Q_1=mL=0.2(3.34x10^5)=66,800 J. Then warm the water: Q_2=mc Delta T=0.2(4200)(20)=16,800 J. Total Q=Q_1+Q_2=83,600 J.

**Final answer: mL+mcDeltaT=83,600 J.**